Week 08: Robustness: Adversarial Examples, Poisoning and Backdoors

Machine Ethics and Artificial Intelligence Safety

Machine Ethics and Artificial Intelligence Safety (11118BLG001). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Small, deliberate changes to inputs or training data can make accurate models fail. This week covers adversarial examples, attacks at training time, the author's work on adversarial threats to AI safety and the practice of evaluating defences honestly [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to derive and implement the fast gradient sign method and projected gradient descent. They should be able to explain adversarial training as robust optimisation and to describe poisoning and backdoor attacks. Finally, they should be able to design an evaluation that states a threat model and uses adaptive attacks.

Lecture

Adversarial examples

Szegedy and colleagues found that imperceptible perturbations could change the predictions of deep networks, and that the same perturbations often transferred between models [1]. Goodfellow, Shlens and Szegedy explained the phenomenon by the approximately linear behaviour of networks in high dimensions [2]. Many tiny changes aligned with the weights add up to a large change in the output. Their fast gradient sign method (FGSM) perturbs an input x with label y as x' = x + eps sign(grad_x L(x, y)). Here L is the training loss, and eps bounds the change of each input component.

Madry and colleagues framed robustness as a min-max problem: Train the parameters to minimise the loss under the worst perturbation within the allowed set [4]. The inner maximisation is approximated by projected gradient descent (PGD), which takes several small gradient steps and projects back into the allowed set after each one. Carlini and Wagner designed optimisation-based attacks that defeated defences previously thought robust, which made strong attacks the standard for evaluation [5]. Eykholt and colleagues showed that the

threat reaches the physical world: Stickers on stop signs caused misclassification under varying viewing conditions [6].

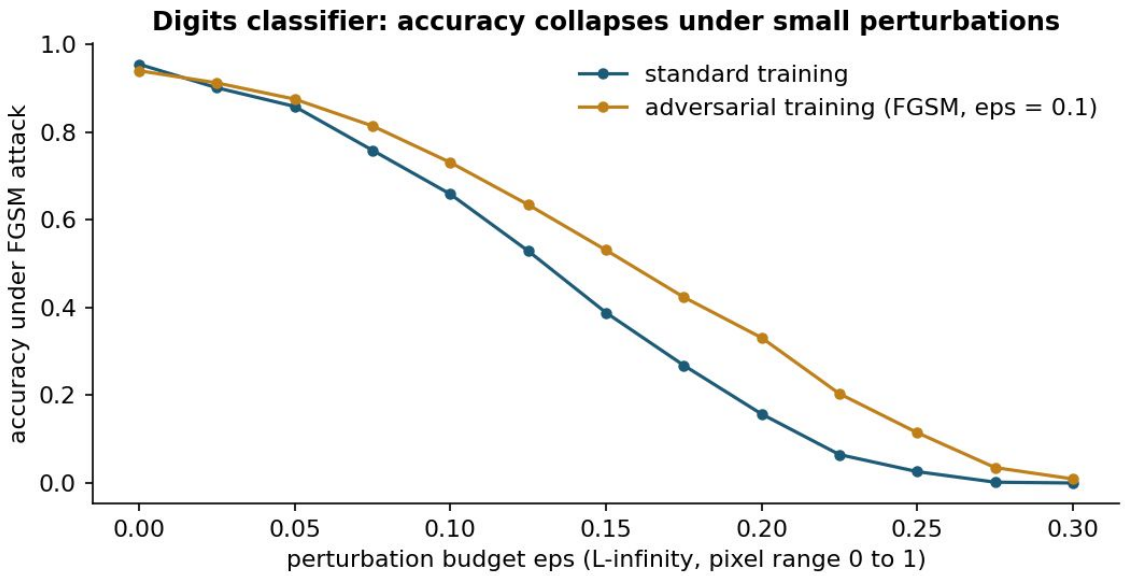


Figure 8.1. Accuracy of a softmax classifier on the digits data set under FGSM attacks of growing budget, with standard and adversarial training [2, 4].

Adversarial threats to AI safety

Kose reviewed techniques for generating adversarial examples and argued that they threaten the safety of AI-based systems in general, not only their security [3]. Kose and Deperlioglu examined adversarial examples as a threat in biomedical engineering problems, where a manipulated signal or image can alter a diagnostic decision without any visible sign [7]. Kose, Cankaya and Deperlioglu discussed a broader future in which AI is both a weapon and a target in cyber conflict [8].

Biggio and Roli traced adversarial machine learning back to attacks on spam filters and described the field as an arms race between attacks and defences [9]. Their central methodological lesson is the threat model. An evaluation must state the attacker's goal and the attacker's knowledge of the system. It must also state the attacker's capability, such as the size of allowed perturbations or the fraction of training data that can be modified.

Poisoning and backdoors

Attacks at training time modify the data or the model before deployment. Poisoning attacks insert crafted examples to degrade accuracy or to cause targeted errors [9]. Backdoor attacks are more insidious. Gu and colleagues showed with BadNets that a network can be trained to behave normally on clean inputs [10]. The same network produces an output chosen by the attacker whenever a small trigger pattern is present, such as a sticker on a traffic sign. Because the model performs well on standard tests, the backdoor survives ordinary validation.

Outsourced training, pretrained models and scraped data therefore form a supply chain whose integrity matters for safety.

Defences and honest evaluation

Adversarial training with PGD remains one of the most reliable empirical defences, at the cost of extra computation and often some clean accuracy [4]. Many other defences, such as input transformations that hide gradients, failed once attacks were adapted to them [5]. A credible robustness claim therefore reports accuracy as a function of the perturbation budget and uses strong iterative attacks with several restarts. It tries attacks designed against the specific defence and states the threat model explicitly [9].

Figure 8.1 shows the pattern on a simple digits classifier. Accuracy falls quickly as the budget grows, and adversarial training shifts the curve without removing the vulnerability.

> ***Pause and reflect.*** *A hospital plans to deploy an image classifier for skin lesions. Write down a threat model: Who might attack the system, what would they know and what could they change?*

Hands-on components

Interactive lab: Each point is a sample of one of two classes, and the line is the decision boundary of a fixed linear model. Pick a point and a perturbation budget. The square shows every input the attacker may choose, and the arrow shows the worst-case FGSM step. The lower chart shows robust accuracy over the whole data set [2, 4]. Link: https://utkukose.github.io/machine-ethics-ai-safety-NB-lecture/weeks/week-08/lab.html

Colab notebook: It attacks a softmax classifier on the scikit-learn digits data set with FGSM and PGD and visualises the adversarial digits. It then applies adversarial training and plants a small backdoor to measure its success rate [2, 4, 10]. Link: https://colab.research.google.com/github/utkukose/machine-ethics-ai-safety-NB-lecture/blob/main/weeks/week-08/NB08_robustness.ipynb

Reflection prompts

1. State a complete threat model (goal, knowledge, capability) for an AI system in your field.
2. Why can a model be accurate and still unsafe? Use the backdoor experiment in your answer.
3. Which part of the machine learning supply chain in your own projects is least verified, and how could you check it?

Weekly task and submission

What to do. Write a robustness evaluation plan for one model in your field. The plan has four parts: the threat model, the attacks you would run, the metrics and perturbation budgets you would report, and the defence you would try first. The threat model states who attacks, what they know about the system and what they can change.

Where to look. Lecture: sections "Adversarial examples", "Poisoning and backdoors" and "Defences and honest evaluation". Interactive lab: the attack on a linear classifier with a perturbation budget (tab: Explore). Colab notebook: Section 2 for projected gradient descent, Section 3 for adversarial training and Section 4 for the backdoor.

Example. For a skin lesion classifier, the attacker could be someone who edits uploaded photos. The plan could use FGSM as a quick baseline and PGD as the stronger attack, report accuracy at three budgets and try adversarial training first.

What to submit. About 400 words and the Colab notebook with Exercises 1 and 2 completed. Include one accuracy curve from the Colab notebook to show how the results would be reported.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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